Functional comparison of the two VSL tasks

Both tasks study a three-to-two lane drop with ending lane **E**, adjacent continuing lane **A** and far lane **B**. They use continuous traffic mass, a finite initial cohort, one constant common speed advice, and count all traffic's time until exit at the choke. The task-first formulation uses a homogeneous approach and a dedicated merge/recovery model; the peer-backed formulation uses lane-cell dynamics adapted from Subraveti, Han and Carey. Their lane-information/control question is shared; functional equivalence or a performance advantage is not assumed.

Aspect	Original task-first formulation	Peer-backed formulation
Traffic state and propagation	Homogeneous lane densities in the approach; a finite terminal uses cumulative arrivals, a travel delay and point discharge (link-transmission-style).	Lane densities in successive physical cells across the whole road (cell-transmission-type). The current specification uses 100 m cells and 6 s steps.
Early merging	A fixed interaction length sets the E-to-A offer; spare A service limits transfer. With sufficient A service, remaining E density decays exponentially with distance traveled.	Desired changes respond to perceived lane densities, downstream lookahead and route/keep-right/cooperation incentives. Distance-to-end urgency remains; shared service limits realized flow.
Permitted lane changes	Only E-to-A in preparation and at the choke. No ordinary lateral change inside the terminal; B remains separate.	Adjacent $\mathrm{E} \leftrightarrow \mathrm{A}$ and $\mathrm{A} \leftrightarrow \mathrm{B}$ changes are permitted before the end, subject to incentives. At the end E must enter A or wait.
Advice and flow capacity	Advice changes the effective triangular flow–density diagram's critical density and capacity. Undisturbed point-service ceilings stay fixed; merge work and recovery constrain actual discharge.	The nominal diagram stays fixed. Advice caps sending; Han's density-dependent discharge/receiving laws govern congestion and capacity drop. Receiving changes through traffic state.
Forced-merge allocation	Unqueued A gets priority if its arriving traffic fits. When both queue, A/E share a work budget; E can require extra work, including a speed-mismatch cost.	E's whole sending offer requests A service. Fixed equal priorities share the receiver, capped by demand; unused shares are redistributed. Desire does not raise priority. E has no extra work cost per unit.
Queues and storage	Its operating bounds keep the approach free, prevent terminal storage blocking entry, and leave B unqueued. Waiting occurs at the A/E merging point.	Finite storage and receiving constraints apply throughout the road. Any lane can accumulate traffic; congestion can extend upstream across cells.
After-exit effects	Forced E entries retain an age- and insertion-speed-dependent influence after completing their trips. Recent entries reduce service until their downstream effect recovers.	Exited traffic has no separate lingering restriction. Past movements affect service through current lane-cell densities; no post-exit recovery account is retained.
Travel-time objective	Complete travel time equals free travel plus A/E waiting. Merge delay is generated by the service/recovery dynamics, then counted by total system time.	Complete population-time is a finite trajectory sum plus an exact remaining-time value. Total system time adds no special E-merging penalty.
Advice sensitivity	Continuous-time traffic dynamics and the timing of arrival and queue-formation/clearance events must be differentiated together.	Differentiate fixed-step stock updates and the exact remaining-time value while active rules stay fixed; treat switching points separately.
Hidden lane information	E and total A+B can conceal an A/B imbalance. With homogeneous initialization, adding initial A mass reconstructs the full initial state.	E and A+B are observed at each section. One extra final-A density identifies the two diagnostic states; broader fields may require more information.